

Cybersecurity & Networking Assignment

1. Which of the following is NOT a component of the CPU?

Answer: RAM is not a component of CPU.

2. What is the function of RAM in a computer?

Answer: RAM stores temporary data and helps the computer run programs quickly.

3. Which of the following is a primary storage device?

Answer: HDD and SSD are primary storage devices.

4. What is the purpose of a GPU?

Answer: GPU is used for graphics, gaming, and video rendering.

5. True or False: The motherboard is the main circuit board of a computer.

Answer: True.

6. Explain the difference between HDD and SSD.

Answer: HDD uses magnetic disks while SSD uses flash memory. SSD is faster than HDD.

7. Describe the function of BIOS in a computer system.

Answer: BIOS starts the computer and checks hardware during booting.

8. List and briefly explain three input devices.

Answer: Keyboard is used for typing, mouse is used for pointing, and scanner is used for scanning documents.

9. Demonstrate how to install a RAM module into a computer.

Answer: Open the RAM slot, align the RAM correctly, and press it gently until it locks.

10. Discuss the importance of cooling mechanisms.

Answer: Cooling prevents overheating and improves computer performance and hardware life.

11. What precaution should be taken before working on hardware?

Answer: Wear an anti-static wrist strap to avoid electrostatic damage.

12. What is the purpose of thermal paste?

Answer: Thermal paste improves heat transfer between CPU and heat sink.

13. Which tool is used to measure PSU voltage?

Answer: A multimeter is used to measure PSU voltage.

14. Which component stores BIOS settings?

Answer: CMOS battery stores BIOS settings.

15. True or False: It is essential to format a new hard drive before use.

Answer: True.

16. What is RAID?

Answer: RAID is a technology used for data storage, backup, and performance improvement.

17. What is the primary function of a router?

Answer: Router forwards data packets between networks.

18. What is the purpose of DNS?

Answer: DNS converts domain names into IP addresses.

19. What type of network topology uses a centralized switch?

Answer: Star topology uses a centralized switch.

20. Explain the difference between a hub and a switch.

Answer: Hub sends data to all devices, while switch sends data only to the correct device.

21. Describe troubleshooting steps for network connectivity.

Answer: Check cables, restart router, verify IP settings, and test with ping command.

22. What is the first step in troubleshooting?

Answer: The first step is identifying the problem.

23. Which utility is used to view system logs in Windows?

Answer: Event Viewer is used to view system logs.

24. What is Safe Mode?

Answer: Safe Mode is a diagnostic mode used for troubleshooting Windows problems.

25. What is the purpose of DHCP?

Answer: DHCP automatically assigns IP addresses to devices.

26. What is a firewall?

Answer: Firewall monitors and controls incoming and outgoing network traffic.

27. What is VPN?

Answer: VPN creates a secure encrypted connection over the internet.

28. Explain TCP/IP networking model.

Answer: TCP/IP model includes Application, Transport, Internet, and Network Access layers.

29. Explain LAN and WAN network.

Answer: LAN covers a small area while WAN covers a large geographical area.

30. Explain operation of switch.

Answer: Switch transfers data using MAC addresses.

31. Explain switch boot sequence.

Answer: Switch boot sequence starts POST, loads IOS, and loads startup configuration.

32. Explain three methods to access switch CLI.

Answer: Console, Telnet, and SSH are methods to access switch CLI.

33. What is Active Directory?

Answer: Active Directory is Microsoft's directory service used for managing users and computers.

34. What is LDAP?

Answer: LDAP stands for Lightweight Directory Access Protocol.

35. What is Linux?

Answer: Linux is an open-source operating system.

36. Explain chmod command.

Answer: chmod command changes file and folder permissions in Linux.

37. How to check Linux memory utilization?

Answer: Use free -m command to check memory usage.

38. What is shell?

Answer: Shell is a command interpreter used to interact with Linux.

39. What is Bash?

Answer: Bash is a popular Linux shell.

40. What is cloud computing?

Answer: Cloud computing means using internet-based services for storage and computing.

41. Describe cloud deployment models.

Answer: Cloud deployment models are Public, Private, and Hybrid cloud.

42. What is cyber security?

Answer: Cyber security protects systems and data from cyber attacks.

43. What are the objectives of cyber security?

Answer: Main objectives are confidentiality, integrity, and availability.

44. Explain offensive and defensive security.

Answer: Offensive security tests systems while defensive security protects systems.

45. Explain types of hackers.

Answer: Types of hackers are White Hat, Black Hat, and Grey Hat hackers.

46. What is SOC in cyber security?

Answer: SOC stands for Security Operations Center.

47. Explain HTTP and HTTPS.

Answer: HTTP is unsecured while HTTPS provides secure communication.

48. What is SSL and TLS?

Answer: SSL and TLS are security protocols used for encrypted communication.

49. Explain password attacks.

Answer: Password attacks include brute force, dictionary, and phishing attacks.

50. Explain SQL Injection.

Answer: SQL Injection is an attack where malicious SQL queries are inserted into input fields.

51. Explain difference between VA and PT.

Answer: VA identifies vulnerabilities while PT tests and exploits vulnerabilities.

52. What is IDS and IPS?

Answer: IDS detects attacks while IPS detects and blocks attacks.

53. Explain symmetric and asymmetric cryptography.

Answer: Symmetric uses one key while asymmetric uses public and private keys.

54. Describe VPN and IPSec.

Answer: VPN provides secure remote access while IPSec secures IP communication.